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Sparse MoE Routing

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for Long Contexts

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Abstract. We study top-@Mk gating in mixture-of-experts (MoE) transformers as the context window grows from 8K to 128K tokens. At long contexts the router becomes the bottleneck: load imbalance grows super-linearly with sequence length and routing entropy collapses. We propose *entropy-anchored routing*, a constant-overhead modification that maintains expert utilisation within 8% of uniform across all context lengths we tested, while improving perplexity by 0.34 nats on the PG-19 long-document benchmark.

1. Introduction

Long-context language models stress every subsystem of a transformer, but the routing layer in mixture-of-experts variants is the first to break. At a context of 128K tokens the per-token routing cost is amortised across many more tokens than at 8K, yet the load distribution across experts is observed to grow markedly more uneven. The standard top-@M{k} gate of Shazeer et al. [1] balances load via an auxiliary loss that pulls gating logits toward uniform; at long context this auxiliary loss is dominated by the cross-entropy term and contributes negligibly to gradients.

The contribution of this poster is a single drop-in change to the gate that reintroduces an explicit entropy floor. We call this scheme @I { entropy-anchored routing }. The modification adds 0.4% to the forward-pass FLOPs and no parameters. We evaluate on three open MoE checkpoints (Mixtral-8x7B, Switch-Base-128, and our own 1.3B-A0.3B model) and on four long-context benchmarks.

2. Method

Let x_t \in \mathbb{R}^d be the token representation at position t and let W_r \in \mathbb{R}^{E \times d} be the router weight matrix for E experts. The standard top-k gate computes

$$g_t \;=\; \mathrm{softmax}\!\bigl(W_r x_t\bigr), \qquad y_t \;=\; \sum_{e \in \mathrm{TopK}(g_t)} \frac{g_{t,e}}{\sum_{e' \in \mathrm{TopK}(g_t)} g_{t,e'}} \; f_e(x_t).$$

Our modification replaces the softmax with a tempered softmax whose temperature \tau_t is chosen, per token, to enforce a target entropy H^\star:

$$g_t \;=\; \mathrm{softmax}\!\bigl(W_r x_t / \tau_t\bigr), \qquad \tau_t \;=\; \arg\min_{\tau > 0} \bigl(H(g_t(\tau)) - H^\star \bigr)^2.$$

The one-dimensional minimisation is solved to within 0.01 nats in three Newton steps – the entropy is monotone in \tau, so convergence is immediate. Figure ? sketches the routing pipeline.

[inline SVG omitted in non-SVG back-end]

Figure 1. Routing pipeline with the entropy-anchored gate. The tempered softmax block is the only departure from a vanilla top-k gate; downstream experts are untouched.

3. Results

We compare three gating schemes – vanilla top-2, top-2 with load loss [2], and entropy-anchored – on four context lengths. Across all four lengths the entropy-anchored gate maintains expert utilisation within 8% of uniform; the load-loss baseline drifts to 31% imbalance at 128K. Perplexity on PG-19 improves by 0.34 nats over the load-loss baseline at 128K and by 0.07 nats at 8K, suggesting the benefit grows with context.

[inline SVG omitted in non-SVG back-end]

Figure 2. Expert utilisation imbalance as context grows. Vanilla top-2 degrades fastest; the load-loss term [2] only delays the problem. Entropy anchoring is essentially flat.

context	indent right	**vanilla**	indent ctr	**load-loss**	indent ctr	**anchored**	indent ctr
8K	indent right	4.2%	indent ctr	3.8%	indent ctr	3.7%	indent ctr
16K	indent right	7.1%	indent ctr	5.9%	indent ctr	4.1%	indent ctr
32K	indent right	13.0%	indent ctr	9.4%	indent ctr	5.3%	indent ctr
64K	indent right	22.0%	indent ctr	16.3%	indent ctr	6.8%	indent ctr
128K	indent right	35.0%	indent ctr	31.0%	indent ctr	7.9%	indent ctr

4. Discussion

The result that surprised us most was the @I { robustness } of the anchored scheme to choice of H^\star: any target between 0.6 and 0.9 times \log E gave indistinguishable downstream perplexity, even though the realised temperatures differed by an order of magnitude. This argues that the auxiliary loss in [1] is solving the right problem – enforce entropy – but with the wrong tool, a soft penalty whose strength dissolves at long context.

Two threats to validity. First, our 1.3B model is small enough that gate behaviour may not generalise to the 70B+ scale; we plan a replication at 30B. Second, our long-context evaluations rely on PG-19, which is literary text; routing on code or scientific prose may differ.

5. Conclusion

Entropy-anchored routing is a one-line change to the top-k gate that restores the load-balancing property load-loss methods lose at long context. The overhead is negligible (0.4% forward FLOPs, zero parameters) and the perplexity gain grows with context length. Source and pretrained gates are available at the mdlout group GitHub.

References

References

1. N. Shazeer et al. Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. *ICLR*, 2017.
2. W. Fedus, B. Zoph, N. Shazeer. Switch transformer: scaling to trillion parameter models. *JMLR*, 23(120):1-39, 2022.